

Public Economics

Take-Home Assignment 2

Topics: Cost-Benefit Analysis, Political Economy

Name: _____ Student ID: _____ Due: _____

Answer both questions. Where a short explanation is requested, 2–3 sentences is enough — we are grading for correct reasoning, not length. Show your work for any calculation.

Question 1: Discounting and Cost-Benefit Analysis

The city of Linz is evaluating a new tram line. It costs €40 million today (year 0) and is expected to generate net social benefits of €5 million per year for the next 10 years (years 1–10).

- a) Using a discount rate of 5%, the present value of a €5 million/year benefit stream over 10 years is approximately €38.6 million. Calculate the project's net present value (NPV) at this discount rate. Should the city build the tram line on cost-benefit grounds alone?
- b) A city council member argues the discount rate should instead be 2%, since “future generations' well-being matters just as much as ours.” Without recomputing the exact NPV, explain in 2–3 sentences which direction this change would push the project's NPV, and why. Then explain, in one sentence, why the choice of discount rate is often the most contested part of a real-world cost-benefit analysis.

Question 2: The Median Voter

Three residents of a small Austrian village must decide, by pairwise majority vote, how large a new public swimming pool should be (measured in square meters). Resident A wants 200 m², Resident B wants 500 m², and Resident C wants 900 m². All three have single-peaked preferences: each resident's satisfaction falls the further the actual size is from their own ideal.

- a) According to the median voter theorem, which pool size wins against any other proposal put to a majority vote? Explain briefly why, in terms of what happens when someone proposes a size above or below it.
- b) A fourth resident, D, moves into the village. D's preferences are also single-peaked, but D wants a pool of just 50 m² (preferring to spend the money elsewhere). Does the outcome that wins under majority vote change? Explain your reasoning.